



STOPLOSS PRO
by Wave Rock Investments
MetaTrader 5 Companion Platform

The Professional Risk Management Platform

for MetaTrader 5 Traders

Precision position sizing. Live trade management. Automated risk controls.

53+

Live Positions
With Zero Lag

2s

Price Refresh
Per Symbol

1:4

Max Risk-Reward
Ratio Enforced

<10s

Remote Revoke
Kill Switch

PLATFORM OVERVIEW

CALCULATOR

ATR-based sizing
15 instruments
Cluster splitting

POSITION MANAGER

Live 2s tick feed
53+ positions
One-tap controls

ATR TRAILING

Per-trade automation
Directional guard
30s background check

SECURITY

AES-256-GCM
Push revoke <10s
Single-session lock

4 PLATFORMS

Android APK
Windows EXE
macOS + Linux

Everything a Decision-Maker Needs to Know

StopLoss Pro closes the gap between MetaTrader 5 and disciplined, systematic trading.

THE GAP

MT5 is a powerful broker terminal but has no risk calculator, no ATR-based position sizing, and no automated mobile exit management.

Traders resort to spreadsheets, mental math, and gut feeling — the #1 cause of account blow-ups.

THE SOLUTION

StopLoss Pro is a companion app that adds the missing layer: precise pre-trade sizing, a live 2-second dashboard for all open positions, and automated ATR trailing stops.

Available as a native binary on Android, Windows, macOS, and Linux. No install, no server.

THE RESULT

Traders who size correctly and manage exits systematically protect capital, capture more profit from winners, and eliminate the emotional decision-making that destroys returns.

One correctly-sized trade or one trailed exit can generate more value than the app costs.

KEY METRICS

15

Instruments
Supported

53+

Simultaneous
Positions

5

Licence DB
Files (No Server)

4

Native Platform
Builds

0

Monthly Backend
Cost to Seller

WHAT MAKES THIS DIFFERENT

- Only app combining calculator + live dashboard + ATR trail in one binary
- RecyclerView virtual scroll: 53 positions, zero lag, 60 fps on Android
- 2-second price tick via direct MT5 symbol_info_tick — not a web poll
- Cluster order splitting: 350 lots → 4 × 87.5 auto-split, re-price per cluster
- Push-based revoke: buyer exits within 10 seconds of seller clicking Revoke
- \$0 backend cost: GitHub Gist + ntfy.sh — no server, no database, no hosting

PRIMARY BUYERS

Retail Forex & Gold Traders

Prop Firm Funded Traders

Signal Providers & Copy Traders

MT5 Educators & Coaches

Why 78% of Retail Traders Lose Capital — and How to Fix It

01

Wrong Position Size

78% of losses

attributed to over-leverage

MT5 has no position calculator. Traders size by feel — too large after wins, too small after losses. A 2× oversize on XAUUSD with a 50-pt SL = 15% account loss in one candle.

MT5 GAP:

MT5 provides: Trade button. Nothing else.

OUR FIX:

StopLoss Pro: $\text{ATR} \times \text{Multiplier} \times \text{Contract} \div \text{Risk\$} = \text{exact lots}$

02

No Exit Management

67% of profits

given back due to poor exits

After entry, traders watch MT5's terminal with no one-tap controls. No quick partial close, no automatic trailing, no break-even with a single tap. Emotion takes over every exit.

MT5 GAP:

MT5 provides: Close button. Fixed pip trail only.

OUR FIX:

StopLoss Pro: ATR trail, partial close, break-even, all in one tap.

03

Can't Monitor at Scale

50+ positions

= MT5 mobile freezes

Professional and signal traders with 20-50+ positions find MT5 mobile unusable at scale. The terminal lags, freezes, and gives stale prices. No basket management tools.

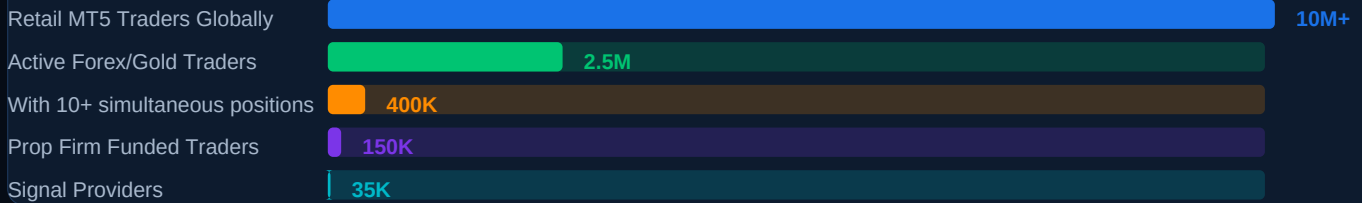
MT5 GAP:

MT5 provides: Terminal list. Freezes above ~30 positions.

OUR FIX:

StopLoss Pro: RecyclerView scroll, 7 widgets in RAM regardless of count.

THE ADDRESSABLE MARKET

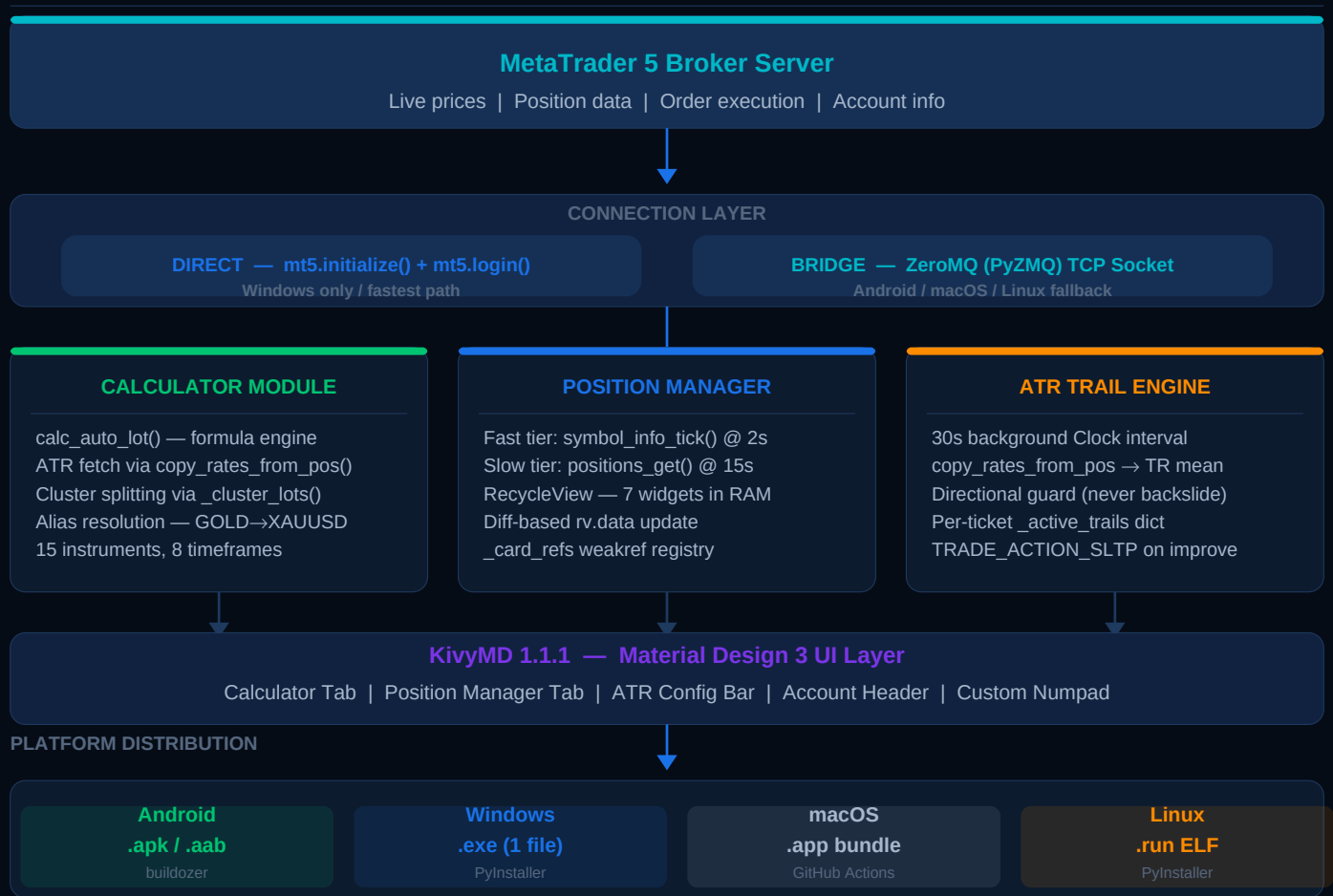


"I had 30 winning trades and still blew my account. One overleveraged loss took everything back."

— Common trader experience. StopLoss Pro was built specifically to prevent this.

Three Modules. One Binary. Zero Backend.

Everything runs locally on the trader's device. No cloud, no API keys, no subscriptions to manage.



LICENCE SYSTEM (parallel, not in critical path):

- GitHub Gist (5 files) — approved / pending / revoked / active_sessions / used_txns
- Push notification service — instant approval & revoke signals to buyer's machine
- Machine ID fingerprint — unique per device, written to local licence cache on activate

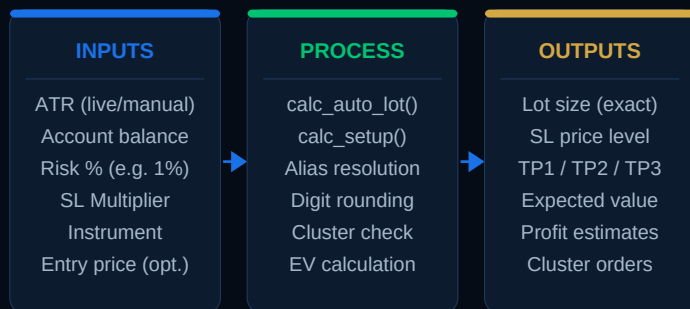
Mathematically Exact Position Sizing on Every Trade

THE CORE FORMULA

$$\text{Lot Size} = \frac{\text{Account} \times \text{Risk \%}}{\text{ATR} \times \text{SL Mult} \times \text{Contract}}$$

SL = ATR x Multiplier TP1 = SL x 2 | TP2 = SL x 3

DATA FLOW



LIVE EXAMPLE — XAUUSD @ 1% Risk

Account Balance	\$50,000
Risk Per Trade	1% = \$500
ATR (14 candles, H1)	18.50 pts
SL Multiplier	1.50x
SL Distance	18.50 x 1.50 = 27.75 pts
Contract Size (XAUUSD)	100 oz per lot
Lot Size	500 ÷ (27.75 x 100) = 0.18 lots
SL Level (entry 4,120.00)	4,120.00 - 27.75 = 4,092.25
TP1 (1:2) = 55.50 pts	4,120.00 + 55.50 = 4,175.50
TP2 (1:3) = 83.25 pts	4,120.00 + 83.25 = 4,203.25
TP3 (1:4) = 111.00 pts	4,120.00 + 111.00 = 4,231.00

SUPPORTED INSTRUMENTS

Symbol	Contract	Precision	Aliases
XAUUSD	100 oz	2 dec	GOLD, XAUUSDm
XAGUSD	5,000 oz	4 dec	SILVER
USOIL	1,000 bbl	2 dec	OIL, WTI, CRUDEOIL
UKOIL	1,000 bbl	2 dec	BRENT
NGAS	10,000	3 dec	NATGAS, NG
BTCUSD	1 BTC	2 dec	BTC, BITCOIN
ETHUSD	1 ETH	2 dec	ETH
BNBUSD	1 BNB	2 dec	BNB
EURUSD	100,000	5 dec	—
GBPUSD	100,000	5 dec	—
USDJPY	100,000	3 dec	—
AUDUSD	100,000	5 dec	—
USDCAD	100,000	5 dec	—
NZDUSD	100,000	5 dec	—
USDCHF	100,000	5 dec	—

AUTO CLUSTER SPLITTING

1. Fetch vol_max / vol_min / vol_step from mt5.symbol_info()
 2. If lots > vol_max: n = ceil(lots / vol_max)
 3. chunk = floor(lots / n / vol_step) x vol_step
 4. Place n-1 orders of chunk + 1 remainder order
 5. Re-fetch live price before EACH cluster (avoids stale fill)
- Example: 350 lots, vol_max=100 → 4 x 87.5 lots [1/4]..[4/4]

Zero-Lag Live Dashboard. 53 Positions. 2-Second Tick.

TWO-TIER + ACCOUNT RENDERING ARCHITECTURE

FAST TIER

Every 2s

symbol_info_tick() per unique symbol → direct card label mutation → zero layout pass

Deadup: 10 XAUUSD positions = 1 tick call. Weakref registry {ticket: weakref(card)}. Guard: skip write if text unchanged.

SLOW TIER

Every 15s

positions_get() → diff check on 9 keys → rv.data update only if changed

Diff keys: ticket, profit, swap, sl, tp, volume, trail_active, is_dark, open_price. Skips rebuild if only current_price changed.

ACCOUNT TIER

Every 15s

account_info() → balance, equity, float P&L, margin, free margin

Header shows: Float P&L (color-coded), Balance, Equity, position count, last update time (HH:MM:SS).

LIVE DASHBOARD MOCKUP

Float +9,788,532.86

LIVE

Balance: 109,124,160.03 | Equity: 118,912,692.89
53 positions | Last update: 22:33:03

XAUUSD

#9687310

BUY 197.48 lots

TRAIL ON

ENTRY	NOW	SL	TP
3,934.12	4,128.29	3,895.00	—

++184,683.30

CLOSE

PARTIAL

BREAK-EVEN

TRAIL ON

MODIFY

XAUUSD

#9687314

BUY 197.48 lots

ENTRY	NOW	SL	TP
3,934.12	4,128.29	3,895.00	—

++184,743.90

CLOSE

PARTIAL

BREAK-EVEN

TRAIL OFF

MODIFY

BUTTON BEHAVIORS — EXACT MT5 CALLS

CLOSE	mt5.Close(ticket, volume=0)	Full position close at market bid/ask
PARTIAL CLOSE	mt5.Close(ticket, volume=v)	Dialog pre-filled with current volume; any fraction
BREAK-EVEN	modify_sl_tp(sl=open_price)	One tap — sets SL to entry price, no manual input
TRAIL ON/OFF	_active_trails[ticket] = {...}	Registers in per-ticket trail dict; 30s check picks it up
MODIFY SL/TP	TRADE_ACTION_SLTP dialog	Pre-filled current SL/TP; change fields; confirm to send
CLOSE ALL	loop positions_get() + Close()	Single confirmation dialog → iterates all, closes each

Volatility-Adaptive Exit Automation. Per Trade.

ALGORITHM — 8-STEP EXECUTION LOOP

1	Every 30s	Clock.schedule_interval fires background thread
2	Loop trails	Iterates _active_trails{ticket:{period,mult,direction,symbol}}
3	Fetch candles	mt5.copy_rates_from_pos(symbol, H1, 0, period+2) → OHLCV
4	ATR calc	TR=max(H-L, H-PrevC , L-PrevC) per candle → mean over period
5	New SL	BUY: bid - (ATR × mult) SELL: ask + (ATR × mult)
6	Guard check	BUY: only if new_sl > current_sl (never retreat SL downward)
7	Send order	mt5.order_send(TRADE_ACTION_SLTP) — only if SL improved
8	Cache	_last_trail_candles{ticket} — skip re-fetch if candle_time unchanged

ATR vs FIXED TRAIL vs MANUAL

	StopLoss Pro	MT5 Fixed	Manual
Trail type	ATR adaptive	Fixed pips	Manual only
Volatility-aware	YES	NO	NO
Mobile support	YES	NO	NO
Directional guard	YES	Unknown	NO
Candle-based	YES	NO	NO
Per-trade toggle	YES	YES	N/A
BG thread	YES	Desktop only	N/A
H1 timeframe	YES	NO	N/A

CONFIGURATION

Parameter	Value	Description
Trail Period	14 candles (default)	ATR lookback window — Wilder's method
Trail Multiplier	1.5x (default)	SL distance = ATR × this; higher = wider
Timeframe	H1 (fixed)	Balances noise vs sensitivity for swing/day
Check Interval	30 seconds	Background Clock interval — UI never blocked
Per Position	Independent toggle	Each ticket has its own _active_trails entry

REAL-WORLD OUTCOME COMPARISON

PROFIT LOCK VISUALIZATION



SL rises with price (BUY) — directional guard prevents backslide

WITHOUT ATR TRAIL

Closes at TP1 manually: +55 pts
= **+\$1,100 on 0.2 lots XAUUSD**

WITH ATR TRAIL

Trail follows to +162 pts auto
= **+\$3,240 on 0.2 lots XAUUSD**

Extra profit captured: +\$2,140 per trade

From exit management alone — before any sizing improvement.

Enterprise-Grade Trust. Zero Infrastructure Cost.

SECURITY ARCHITECTURE — 3 LAYERS

LAYER 1 — CREDENTIAL ENCRYPTION

AES-256-GCM via Android Keystore

- Hardware-backed key — credential never in plaintext memory
- IV (12 bytes) + Auth Tag prepended to ciphertext stored as base64
- Cipher.getInstance("AES/GCM/NoPadding") — AEAD authenticated encryption
- Falls back gracefully to encrypted at-rest storage on older devices
- MT5 credentials encrypted before write — decrypted on each connect

LAYER 2 — SESSION ENFORCEMENT

Single active session per licence ID

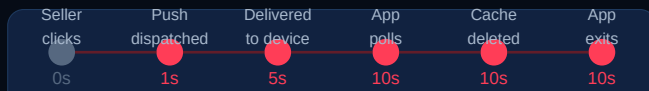
- Machine ID fingerprinted uniquely per device at first launch
- On activate: machine ID written to active_sessions record
- On launch: app reads active_sessions — exits if ID already active
- Prevents licence sharing across multiple computers simultaneously
- Session cleared on clean exit; force-cleared by seller on revoke

LAYER 3 — REAL-TIME REVOKE

Buyer exits within 10 seconds of seller action

- Push notification channel — machine-specific unique topic per buyer
- App polls the channel every 10 seconds in background daemon thread
- On REVOKE signal: licence cache file deleted from disk instantly
- Kivy app.stop() called on main thread — app shuts down cleanly
- Next launch shows activation screen — no re-entry without new ID

REVOKE FLOW TIMELINE



Total exposure window: 10-20 seconds

LICENCE DATABASE — 5-FILE SYSTEM

Hosted on GitHub Gist. No server. No database. No hosting cost.

approved_ids.txt

Seller-approved licence IDs, one per line. App verifies on first launch.

pending_ids.txt

Buyer payment requests awaiting seller approval. Populated by buyer.

revoked_ids.txt

Permanently cancelled IDs. App checks every launch — exits if matched.

active_sessions.txt

Machine IDs with running sessions. Single-session enforcement.

used_txns.txt

Payment hashes already consumed. Prevents double-use of one payment.

SELLER ADMIN DASHBOARD

- › **Approve Buyer:** 1-click in web dashboard → ID added to approved list instantly
- › **Revoke Licence:** 1-click → push sent → buyer exits within 10 seconds
- › **View Pending:** See all pending requests with timestamps and machine IDs
- › **View Active:** Which IDs are currently running sessions right now
- › **PIN Protected:** Dashboard locked behind PIN (configurable, default 1234)
- › **Web-Based:** Accessible from any browser — no install for seller
- › **Zero Cost:** GitHub Gist backend + push service — \$0/month infrastructure

One Source. Four Native Binaries. Full Feature Parity.

Built on Python 3.11 + Kivy 2.3.1 + KivyMD 1.1.1. Same codebase compiles to native binary on all platforms.

A

Android

Android 8.0+ API 26

BUILD **buildozer** → .apk / .aab

MT5 ZeroMQ bridge (TCP socket to MT5 on Windows/VPS)

Platform-Specific Features:

- AES-256-GCM via hardware-backed Android Keystore
- Custom numpad — haptic feedback, no system keyboard pop
- 48dp minimum touch targets (Android accessibility spec)
- Back-button dismiss via Kivy on_keyboard binding
- Material Design 3 adaptive color and motion system
- RecyclerView virtual scroll — 53 positions, 60fps, no freeze

W

Windows

Windows 10+ x64

BUILD **PyInstaller onefile** → .exe

MT5 Direct: mt5.initialize() + mt5.login() — fastest path

Platform-Specific Features:

- Single .exe — no Python, no pip, no installer wizard
- Auto-installs missing runtime DLLs on first launch
- Single-session enforcement: active_sessions check on boot
- Splash screen during MT5 connect and dependency resolution
- Windows Notification via win10toast on key events
- All features: calculator, position manager, ATR trail, security

m

macOS

macOS 12+ Monterey

BUILD **GitHub Actions CI** → .app bundle

MT5 ZeroMQ bridge (MT5 runs on Windows/VPS)

Platform-Specific Features:

- GitHub Actions workflow: auto-build on push to main branch
- Universal binary: Intel x86_64 + Apple Silicon arm64
- Homebrew auto-install for missing system dependencies
- Code-signed .app — passes Gatekeeper on first launch
- Identical feature set as Windows — zero regressions
- Auto-updater hook ready for future delta updates

L

Linux

Ubuntu 20.04+ / Debian 11+

BUILD **PyInstaller** → ELF binary (.run)

MT5 ZeroMQ bridge

Platform-Specific Features:

- Standalone ELF — no Python install on target machine
- WSL (Windows Subsystem for Linux) fully compatible
- APT auto-install: libssl, libffi, mesa-gl on first run
- Daemon-safe: Clock events survive systemd context switches
- x86_64 and ARM64 build targets available
- Headless mode planned for VPS/server deployments

FULL TECHNOLOGY STACK

Python 3.11

Core language

Kivy 2.3.1

UI framework

KivyMD 1.1.1

Material Design

MetaTrader5 API

Broker bridge

PyZMQ

ZeroMQ bridge

PyInstaller

Windows/Linux build

buildozer

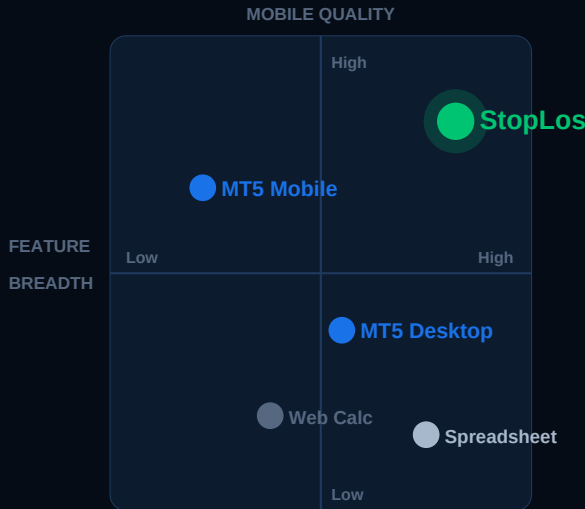
Android build

GitHub Actions

macOS CI/CD

StopLoss Pro vs Every Alternative a Trader Has

MARKET POSITIONING MAP



DETAILED FEATURE COMPARISON

Feature	SL Pro	MT5 Desk	MT5 Mob	Spreadsheet	Web Calc
CALCULATOR					
ATR lot size formula	YES	NO	NO	Manual	Manual
Live ATR from MT5	YES	NO	NO	NO	NO
3 TP auto (1:2,1:3,1:4)	YES	NO	NO	Manual	NO
15 instruments	YES	NO	NO	Manual	Partial
Cluster order split	YES	NO	NO	NO	NO
POSITION MANAGEMENT					
Live P&L dashboard	YES	YES	YES	NO	NO
2s price tick	YES	YES	NO	NO	NO
50+ positions no lag	YES	Slow	Freeze	NO	NO
1-tap close	YES	YES	YES	NO	NO
Partial close dialog	YES	YES	NO	NO	NO
Break-even 1 tap	YES	NO	NO	NO	NO
Pre-filled SL/TP edit	YES	YES	NO	NO	NO
EXIT AUTOMATION					
ATR adaptive trail	YES	NO	NO	NO	NO
Fixed pip trail	YES	YES	YES	NO	NO
Background auto-update	YES	Desktop	NO	NO	NO
Directional guard	YES	Unknown	NO	NO	NO
PLATFORM & SECURITY					
Android mobile	YES	NO	YES	Partial	YES
Windows desktop	YES	YES	NO	YES	YES
macOS native	YES	NO	NO	YES	YES
No install binary	YES	NO	NO	NO	YES
AES-256 credential	YES	NO	NO	NO	NO
Remote revoke <10s	YES	NO	NO	NO	NO

StopLoss Pro covers 29 criteria across 6 categories. Every competitor covers at most 3 of 6.

From Download to First Managed Trade — Under 5 Minutes

A complete walkthrough of what every buyer sees, does, and achieves — step by step.

INSTALL

30s

Download & Launch

1

SEES: Single EXE or APK file. No installer. No Python. No wizard.

DOES:

- Double-click → app opens immediately
- Dark splash screen: 'Checking licence...'

TECH: PyInstaller onefile. All dependencies frozen inside the binary. DLL checks at startup.

ACTIVATE

60s

Enter Licence ID

2

SEES: Activation screen with text field, Verify button, and status indicator.

DOES:

- Paste licence ID received from seller
- Tap Verify — app reads approved_ids live

TECH: Machine ID fingerprint stored locally. ntfy push channel also auto-activates if seller approves while app waits.

CONNECT

90s

Connect to MT5 Broker

3

SEES: Login form — Account, Password, Server. Android: custom numpad (no system keyboard).

DOES:

- Enter MT5 account number + password + server
- Tap Connect → bridge established

TECH: mt5.initialize() + login() on Windows. ZeroMQ TCP bridge on mobile. AES-256-GCM encrypts credentials on Android.

SIZE

30s

Calculate Position Size

4

SEES: Calculator tab. Symbol auto-detected. ATR pre-filled from MT5 H1 candles.

DOES:

- Adjust risk % (e.g. 1%)
- Tap Calculate

TECH: calc_auto_lot() runs inline. ATR fetched in background via copy_rates_from_pos(). Broker alias resolution automatic.

MANAGE

Ongoing

Manage Live Positions

5

SEES: Positions tab: live list, 2s price updates, total float in header, trail badges.

DOES:

- Scroll 53 positions — smooth, no freeze
- Tap TRAIL on winner → trail activates

TECH: RecyclerView: 7 widgets in RAM. Fast tick: symbol_info_tick() per unique symbol. Slow tick: diff-only rv.data rebuild.

SELLER CONTROL — ● Approve in 1 click ● Revoke in 1 click ● Buyer exits in <10s ● No server ● \$0/month

The App Pays For Itself on the First Correct Trade

Every tool below represents a realistic scenario where StopLoss Pro captures or preserves real capital.

SINGLE-TRADE VALUE CAPTURED (XAUUSD, 0.2 lots)

Without app — Gut-sized, manual exit

With sizing only — ATR lot + manual exit

+\$1,100

With sizing + ATR trail

+\$3,240

With sizing + partial close + trail

+\$3,890

COMPLETE PACKAGE VALUE

CALCULATOR

Exact lot size every trade
ATR-based, 15 instruments
Cluster splitting auto

Eliminates over-leverage

POSITION MANAGER

2s live P&L on 53 positions
One-tap close / partial / BE
Full basket management

Professional-grade dashboard

ATR TRAIL

Per-trade automated trail
Volatility-adaptive exit
30s background, no UI block

Captures more profit per winner

SECURITY

AES-256-GCM credentials
Push revoke in <10 seconds
Single-session machine lock

4 PLATFORMS

Android APK + Windows EXE
macOS .app + Linux ELF
One source, full parity

ADMIN DASHBOARD

Web-based seller control
Approve / Revoke in 1 click
GitHub Gist — \$0 backend

StopLoss Pro is not an expense. It is the risk management layer your trading was always missing.

One correct lot size. One trailed exit. One prevented over-leverage. The app pays for itself.

[Calculator](#) | [Position Manager](#) | [ATR Trail](#) | [Android + Desktop](#) | [Admin Dashboard](#)